NEUROSCIENCE

Excessive mobility interrupted

Mobile DNA sequences called L1 contribute to the brain's genetic heterogeneity and may affect neuron function. The protein MeCP2, which is mutated in Rett syndrome, seems to regulate the activity of these genomic elements. SEE LETTER P.443

LORENZ STUDER

In his struggle to rationalize the laws of quantum mechanics, Albert Einstein once wrote that God "does not play dice". A similar struggle comes to mind when trying to understand the biological consequences of several reports from the labs of Gage and Muotri, including one appearing on page 443 of this issue (Muotri *et al.*[1]). These studies offer evidence for the occurrence of quasi-random genetic changes in neurons during development. The work raises provocative questions about whether 'playing dice' during brain development contributes to the differences that make each of us unique, and how such changes may be related to brain disorders.

The immune system is well known for using genetic recombination and random mutations[2] to adapt antibody defences against invaders. The question is, does the nervous system use a similar approach to create neuronal diversity? For example, each olfactory neuron expresses only one of more than 1,000 possible olfactory-receptor genes. Permanent genetic changes are unlikely to underlie selection of olfactory-receptor genes for expression: mice generated from the genetic material of a neuron expressing a single olfactory receptor — by the technique of somatic-cell nuclear transfer — re-establish the full complement of olfactory receptors[3,4]. Such evidence, however, does not rule out the possibility of other permanent genetic changes occurring within the neuronal lineage.

Transposons, or 'jumping genes'[5], can mediate one such mechanism, inducing changes in the DNA. In the human genome, the most common class of transposons is retrotransposons, which have the ability to amplify themselves. For instance, one type of retrotransposon, known as LINE-1 or L1, constitutes a staggering 17% of the human genome. However, only a small fraction of L1 is functionally intact and active, and the role of these retrotransposons in the human genome remains mysterious.

It has been proposed[6,7] that L1 is involved in genome evolution, and there is evidence[8] that retrotransposition can induce genetic changes responsible for human disease. Until recently, L1 activity was thought to be confined to the earliest stages of embryonic development and the germ-cell lineage. But this picture started to crumble a few years ago, when the Gage lab demonstrated[9] active L1 retrotransposition in neuronal precursor cells. Subsequent mechanistic studies identified[10] dual binding sites in the promoter region of L1 for the transcriptional regulators SOX2 and TCF/LEF, which are responsible for changes in the associated chromatin (DNA–protein complexes), and so for the switch between repression and activation of L1 expression (Fig. 1).

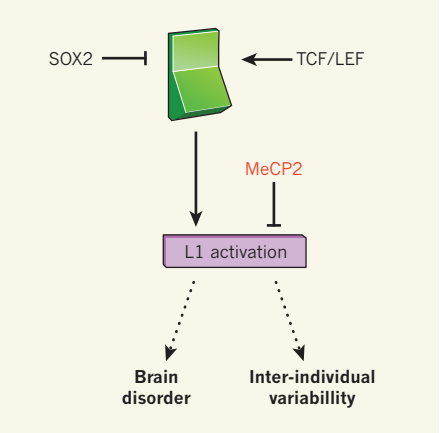

Figure 1 | Regulation of L1 expression. During neuronal development, the expression of L1 retrotransposons is transiently activated when neuronal precursor cells undergo a transcriptional switch from expressing SOX2, which represses L1 expression, to expressing TCF/LEF, which supports it. Muotri *et al.*[1] find that another protein, MeCP2, also regulates L1 expression, repressing its promoter activity and so its retrotransposition rates. Such genetic changes could contribute to brain disorders and variability among individuals.

The intricate regulation of L1 activity was the starting point for the next chapter of this remarkable story, as told by Muotri and colleagues[1]. Previous work in cell lines[11] had suggested that the protein MeCP2 can recruit the enzyme HDAC1 and that the two contribute to L1 repression. A role for MeCP2 in regulating L1 activity was particularly intriguing, because the gene encoding this protein is mutated in Rett syndrome (RTT) — an X-chromosome-linked disorder specific to the nervous system that is a leading genetic cause of mental retardation in girls.

Muotri *et al.* now use a broad armamentarium of approaches to investigate the link between MeCP2 and L1 retrotransposition. They present compelling *in vitro* and *in vivo* evidence that both promoter activity and L1 retrotransposition rates are significantly increased in neuronal precursors of mice that lack the *Mecp2* gene. Indeed, the authors' detailed imaging studies demonstrate clear differences in L1 retrotransposition rates between normal mice and those lacking *Mecp2* across several regions of the adult brain.

Muotri and co-workers also investigate the involvement of MeCP2 in regulating L1 retrotransposition in the cells of patients with RTT. For this, they reprogram the patients' skin fibroblast cells into patient-specific induced pluripotent stem cells (iPSCs), and detect increased L1 retrotransposition rates in neurons derived from the iPSCs of patients with RTT compared with neurons from control iPSCs. The authors further corroborate these data with post-mortem work, finding increased levels of L1 DNA content in the brain, as opposed to matched heart-tissue samples, of patients with RTT, and in brain samples from patients compared with those from age-matched controls.

These findings[1] provide intriguing evidence for the modulation of L1 activity by MeCP2 during development of the nervous system. Is such modulation causally involved in RTT pathology? On the basis of the current knowledge of the disease, probably not.

There is no obvious correlation between the timing of L1 activity during embryonic development and the delayed disease onset in patients with RTT, which typically occurs 1–2 years after birth. It is also difficult to imagine how an increased rate of quasi-random genetic changes, even if biased towards genes expressed in neurons[9], can lead to the highly reproducible disease symptoms observed in RTT. What's more, in *Mecp2*-mutant mice, a striking recovery from RTT symptoms occurs on re-expression of *Mecp2*, even in mature animals — a strategy that clearly does not affect early retrotransposition events. Therefore, changes in L1 activity might not cause RTT, but may contribute to variability among patients — beyond the well-known differences in mutation type and X-chromosome inactivation status.

Another intriguing issue relates to the use of RTT iPSCs in modelling human disease. After reprogramming to iPSCs, the inactive X chromosome of differentiated cells does not become active[12]. Therefore, all differentiated cells generated from a given RTT iPSC line are expected to show identical inactivation patterns of either their normal X chromosome or their *MECP2*-mutant X chromosome, rather than the random inactivation pattern observed in patients. The increased L1 activity in neuronal precursors derived from RTT iPSCs indicates that these cell lines showed inactivation of the normal X chromosome and may, therefore, model a more severe form of RTT.

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The holy grail for defining the functional impact of L1 activation in neuronal development would be a method that can selectively switch on and off retrotransposition events using genetic or pharmacological tools. Given the sheer number and genetic complexity of transposable elements, this remains a daunting task, although targeting the reverse transcriptase enzyme or other mission-critical determinants of L1 activity may represent a potentially tractable approach. Until then, we are left to wonder whether playing dice during development of the central nervous system is indeed part of what makes each of us unique in both health and disease. ■

Lorenz Studer *is in the Center for Stem Cell Biology, Developmental Biology Program, Sloan-Kettering Institute for Cancer Research, New York, New York 10065, USA.*
e-mail: studerl@mskcc.org

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QUANTUM PHYSICS

Entangled quartet

Quantum physics is known for its counter-intuitive principles. One such principle — that a single photon can be in as many as four places at the same time — has now been demonstrated. SEE LETTER P.412

VLADAN VULETIC

When light is shone through two closely separated slits and onto a distant screen, a periodic light pattern emerges as a result of interference between the light waves emanating from the two slits. Where quantum physics is concerned, some of the deepest mysteries — or, in the opinion of the iconic Richard Feynman, the only mystery — arise when that experiment is performed not with strong classical light waves but with a single particle. Although indivisible, a single particle also produces an interference pattern, so it must have passed simultaneously through both slits.

Building on recent advances[1] enabling the storage of single photons in atomic gases, Choi *et al.*[2] (page 412 of this issue) investigate what happens to interference when light is stored simultaneously in as many as four spatially distinct atomic clouds. The authors demonstrate quantum correlations (entanglement) in this composite matter–light system, and study how entanglement ultimately fades away to leave only classical correlations.

Classical correlations can arise in situations in which there is limited knowledge of a system. For instance, if we know only that one coin (or photon) has been hidden in one of four boxes, then detecting the coin in one box would instantaneously tell us that the other three boxes are empty — even if they were separated from each other by light years. It is hardly surprising that such 'particle-type' detection (Fig. 1a) can reveal classical correlations between the numbers of coins found in the different boxes if the total number of coins is known a priori.

Classical correlations can also arise between multiple light waves that are combined on partially reflecting mirrors before detection, such that the origin of the detected light is unknown (wave-type detection). For example, if identical light waves have been stored simultaneously in all four boxes — or, for that matter, coins sufficiently small to display quantum, wave-like character — and the light emerging from the boxes is combined through a series of partially reflecting and totally reflective mirrors before reaching four detectors (Fig. 1b), then the outputs of the detectors would vary as the path length between each box and the corresponding first mirror is varied.

In a classical world, something is either a particle or a wave, so a physical system will exhibit correlations either in the particle-type or wave-type detection set-up — but not in both. However, in the quantum world that we live in, it is possible to place, for example, a single photon simultaneously in all boxes such that correlations are observed in both detection set-ups. And this is exactly what Choi *et al.*[2] have done in their experiment.

Choi and colleagues used four atomic ensembles as the storage boxes. Such systems not only can hold the photon, but also can act as highly directional light emitters that can be triggered on demand through the application of a laser pulse[1,3]. The authors measured correlations between the different boxes, either in

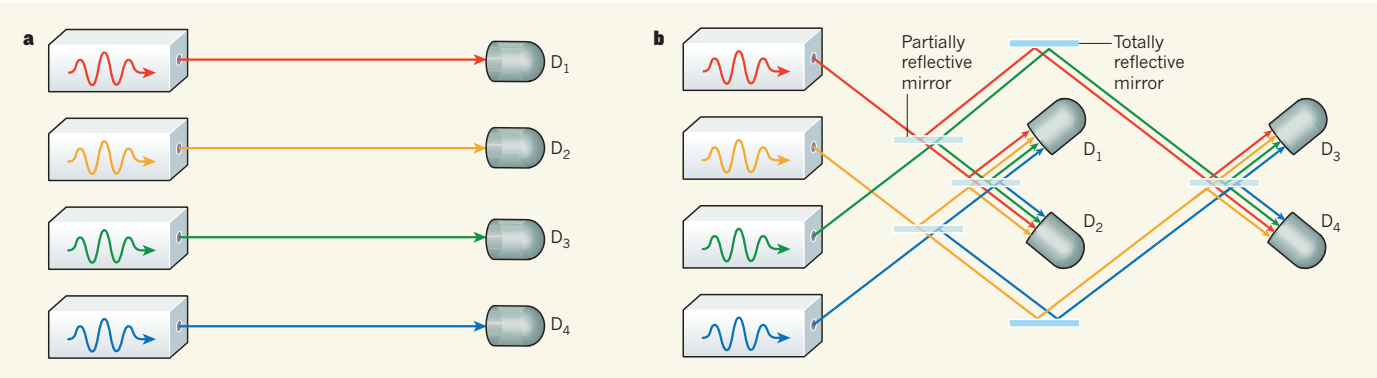

Figure 1 | Particle-type versus wave-type measurements. Choi *et al.*[2] have measured quantum entanglement in a composite matter–light system by combining results from particle-type and wave-type measurements. The matter component of the system consists of four atomic ensembles (illustrated by the boxes) and the light part is a single photon (waveform). **a,** In the particle-type set-up, a photon stored in one box can reach only one detector (D_1, D_2, D_3 or D_4). **b,** In the wave-type measurement, the photon is placed simultaneously in all four boxes and the light emerging from the boxes is combined through an arrangement of partially reflecting and totally reflecting mirrors such that light from any box can reach any detector. The colours and multiple waveforms are for illustration of the photon path only; the light in all four boxes is identical, has the same wavelength, and contains only one photon in total.

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